

Project Management in Innovation and Entrepreneurship

with PONERA case study

EPFL | MSc Course: Project management and risk analysis

06.10.2026

Matthew Reali

EPFL MSc, Co-founder and Managing Director, Ponera Group

EPFL

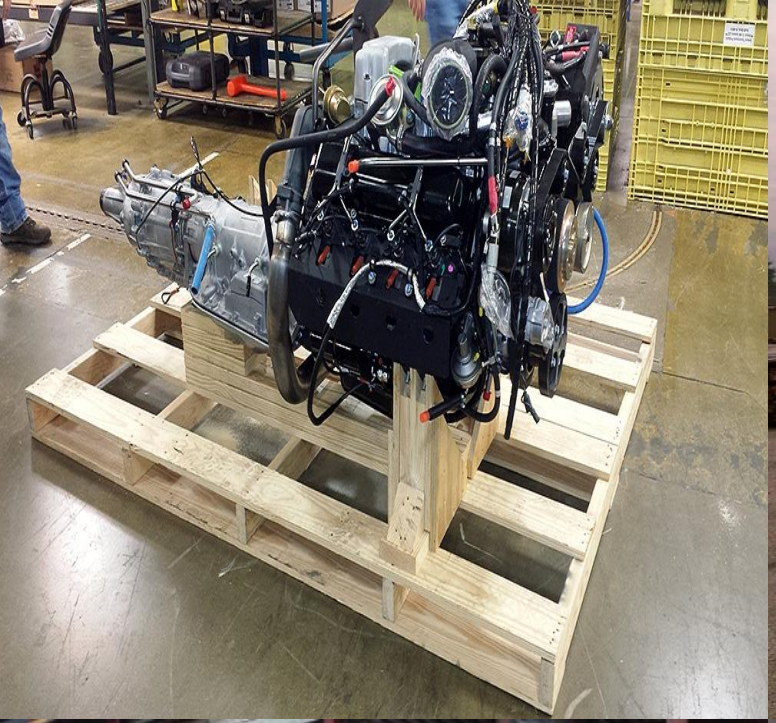


MSc EPFL
MECHANICAL ENG.

3 yrs CREDIT SUISSE
FINANCIAL SERVICES

3 yrs BÜHLER
Operation STRATEGY

5 yrs PONERA
CEO







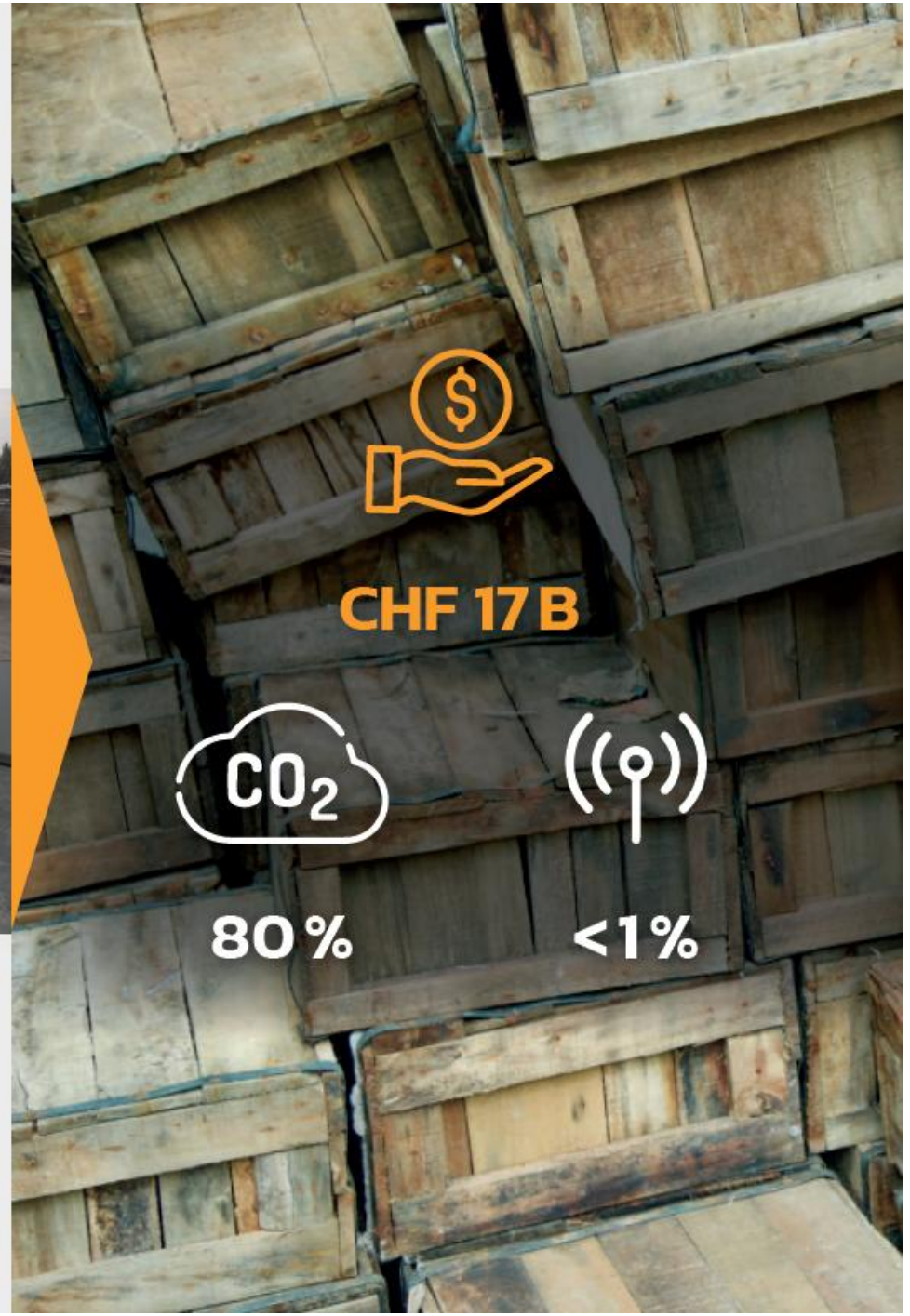






Losing Millions in Industrial Packaging Every Day






30%
Responsible for 30% of
Global Wood Consumption!

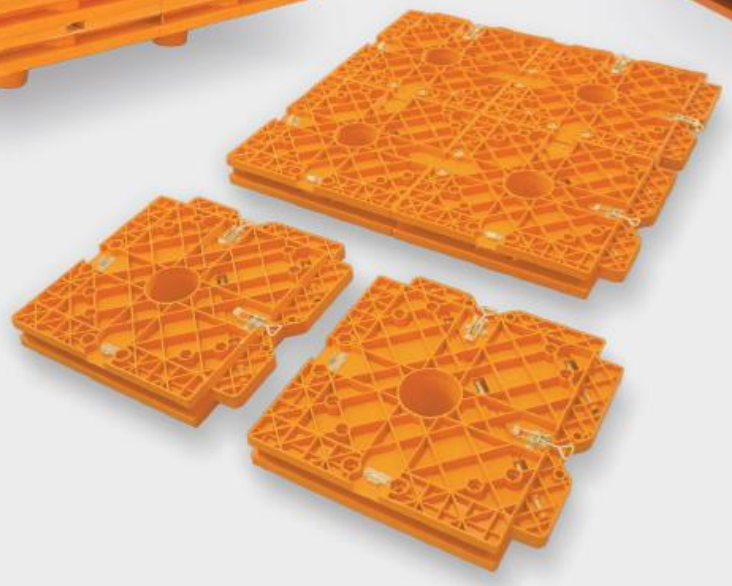


Mission

Ponera Solution Reusable Modular Industrial Packaging







SOLARIMPULSE
EFFICIENT SOLUTION

Huge Reduction in Environmental Footprint



PONERA



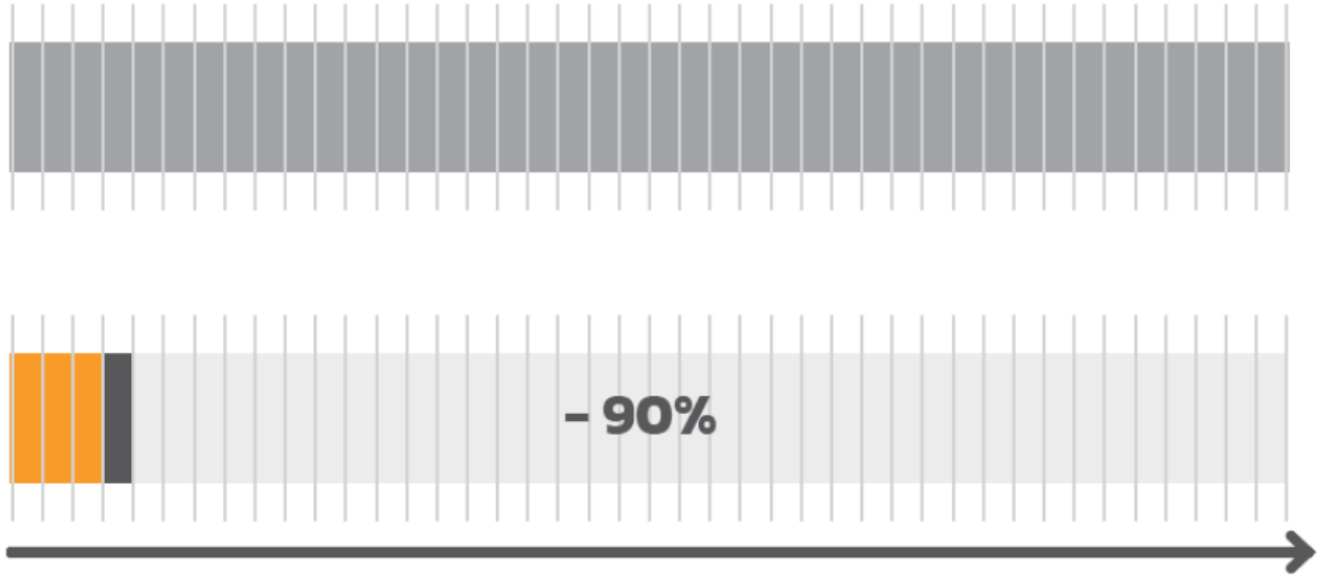
Legno – 95%

CO_2eq – 90 %

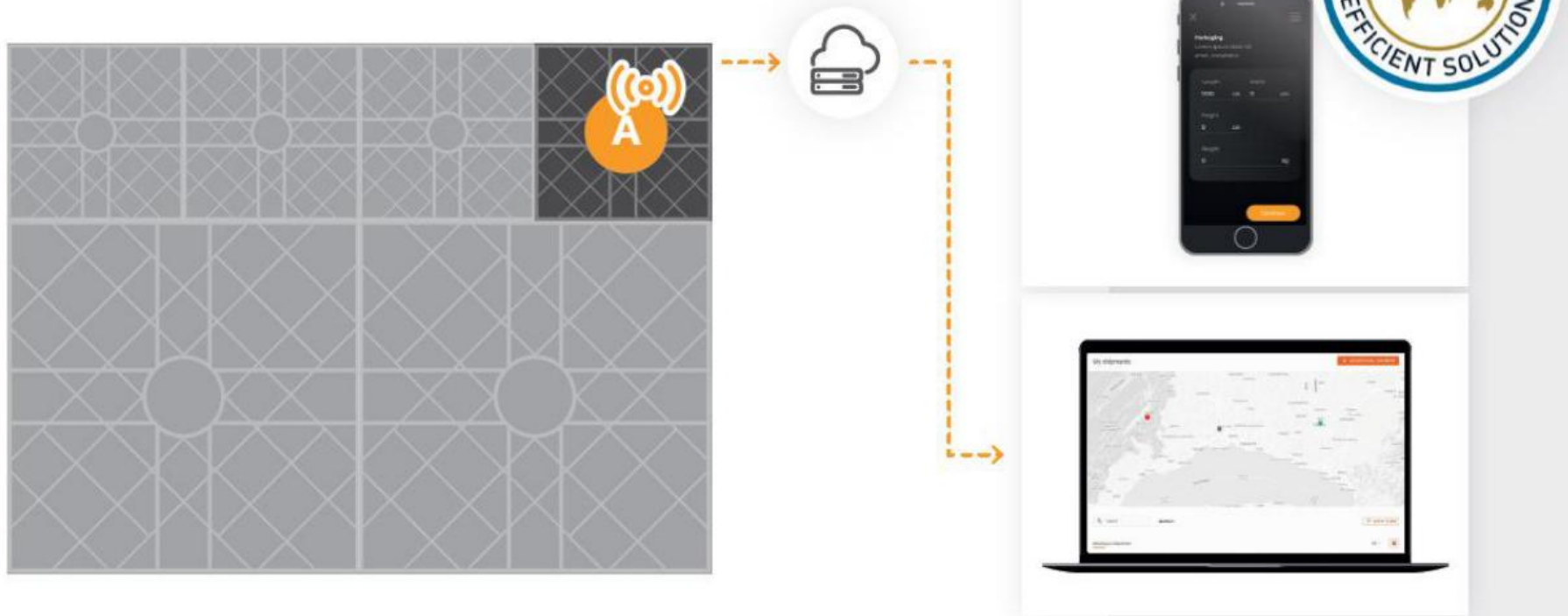
Ponera Offers Huge Cost Savings







Ponera Brings Packaging to the Industry 4.0 Era

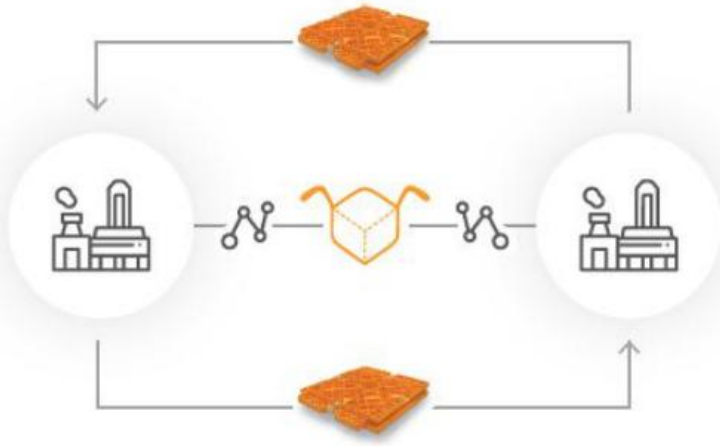


Product Development & Business Model

1. Sales & Leasing Model



+ 2. Activity Based (Data) Services



A Proven Solution











Trasportato prodotti singoli superiori a 15’000 KG!

Current clients

- **7+** companies **$1B-30B** Revenues.
- **15+** companies more than **$10M** in packaging costs

Where (Why) does it Begin?

Two Main Paths to Startup

Start with a real problem or start with **a technology**?

Problem / Product-first startup

Start with a real customer problem and build the best solution



Identify customer problem
A painful, important and widespread need

Validate market
Will customers pay? How big is the market?

Develop solution
Design the best technology and business model to solve it

Build and scale
Industrialize, sell and grow

Key question at the start:
"What important problem can we solve better than existing solutions?"

Advantages
- Clear customer need and willingness to pay
- Higher chance of product-market fit
- Easier to attract talent and investors

Challenges
- Requires deep customer insight and market validation
- May require a different technology than initially expected
- Still hard execution

Typical examples
- Products and services designed around a validated market need
- e.g. modern logistics solutions, fintech for a specific customer pain point, healthcare solutions for a clear unmet need

Technology-first startup

Start with a technology and look for applications and markets



Develop technology
New science, invention or technical innovation

Explore applications
Look for potential use cases and markets

Validate market
Find customers and business model

Build and scale
Productize, industrialize and grow

Key question at the start:
"What can we do with this technology?"

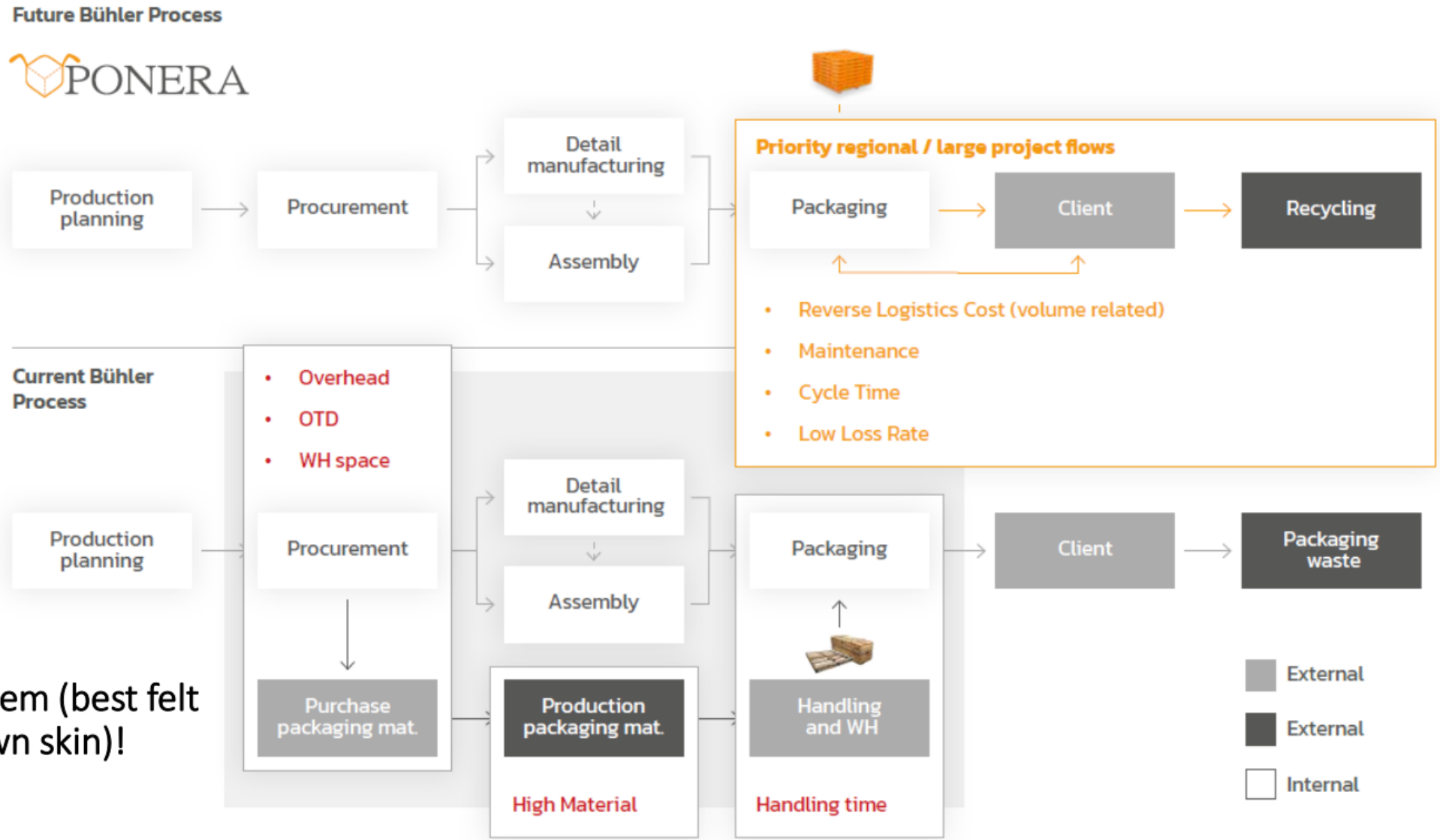
Advantages
- Can create truly disruptive and new markets
- Potential for very high long-term value
- Often driven by strong technical enthusiasm and talent

Challenges
- Risk of looking for a problem that is not important enough
- May take a long time to find a viable market
- Higher risk of "technology looking for a solution"
- Can be difficult to attract customers and investors early on

Typical examples
- Deep tech, new materials, advanced AI, biotech, novel hardware
- e.g. graphene, quantum computing, new battery chemistry, advanced robotics

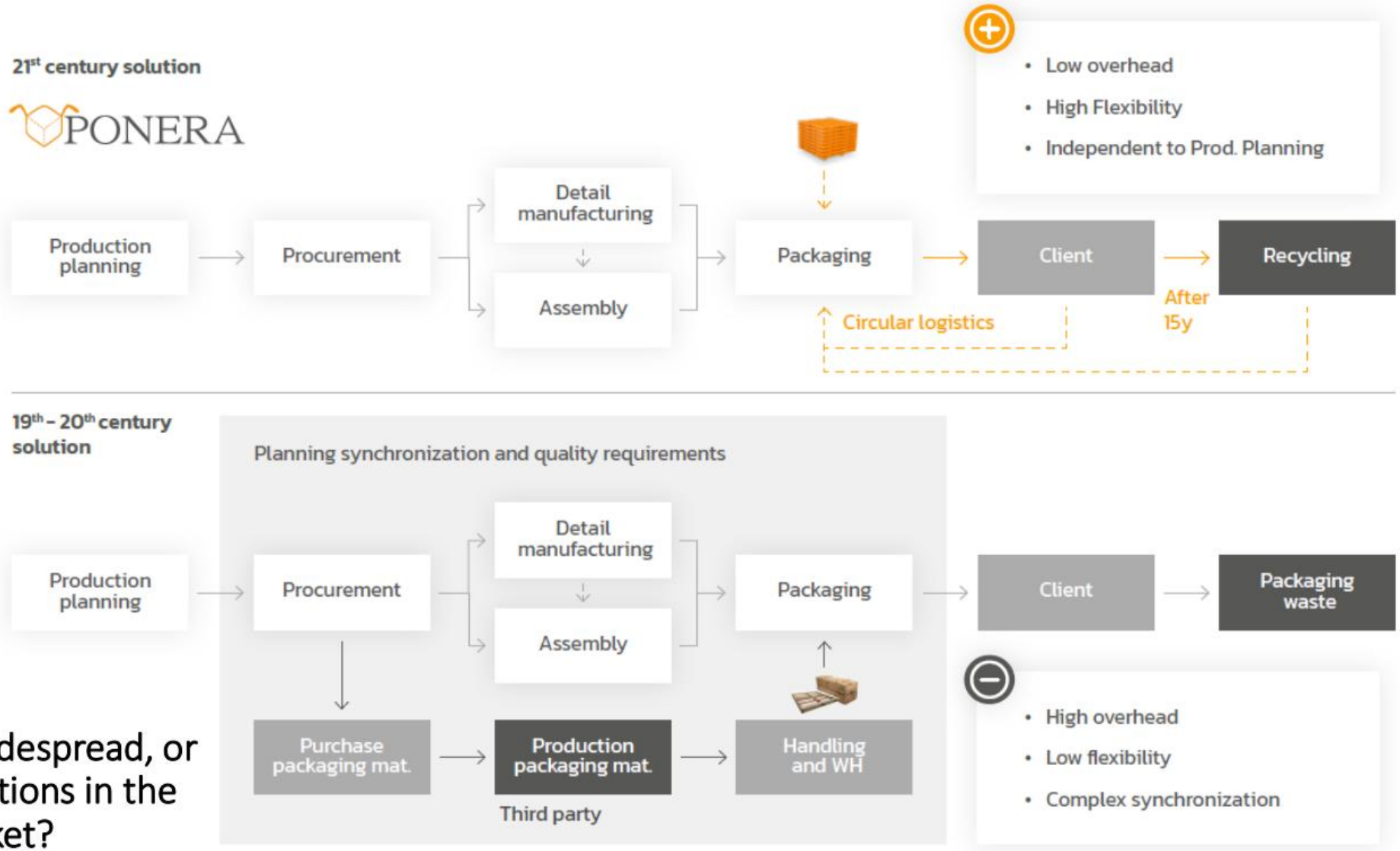
Key takeaway | Both paths can lead to successful companies, but problem-first startups generally have a higher chance of commercial success because they start with a real customer need and validate the market early.

Customer Projects Process Changes



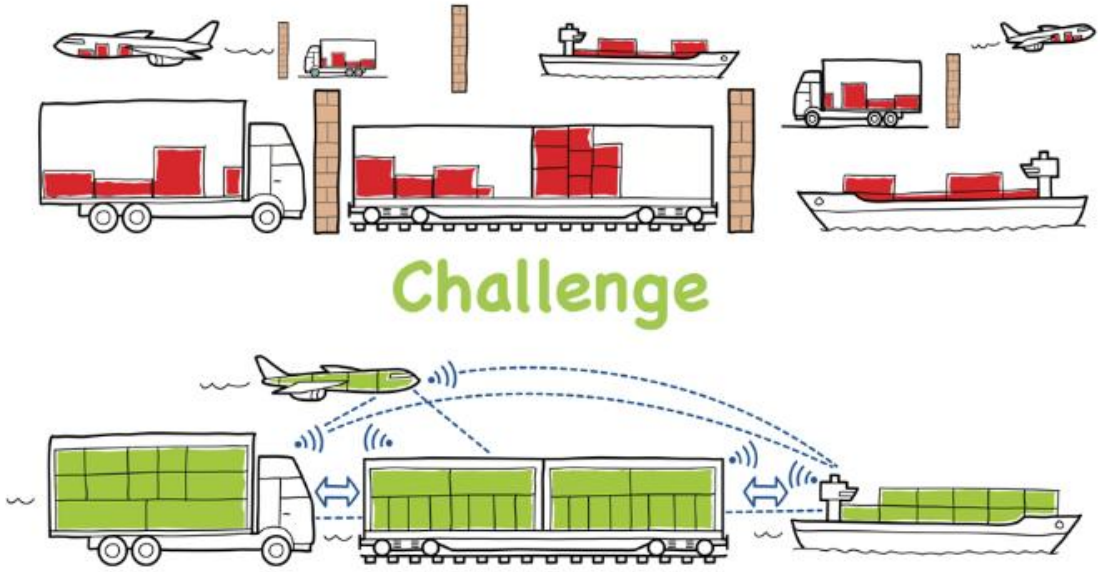
Identify a problem (best felt on your own skin)!

We make life simpler



Is problem widespread, or existing solutions in the market?

Today’s Logistics:



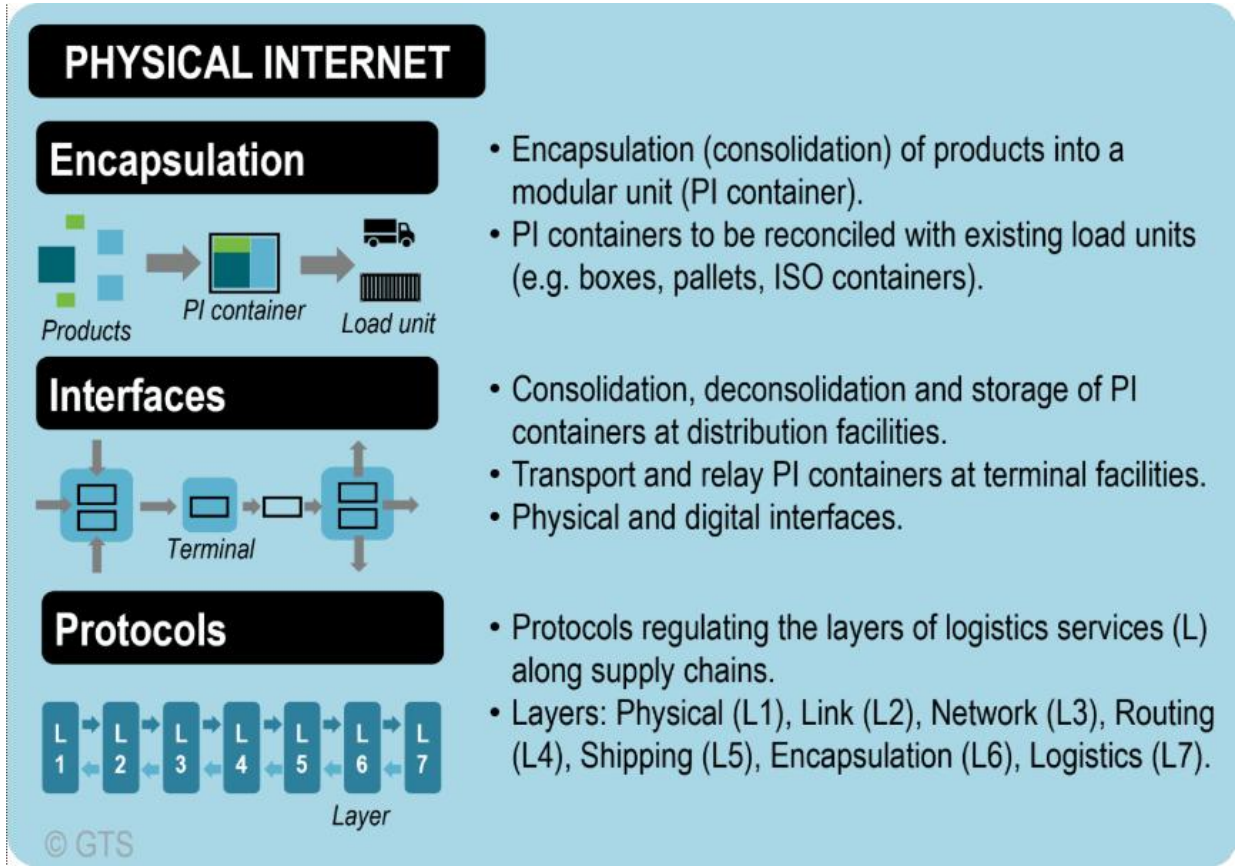
- **50% of Trucks in EU are empty.**
- **Up to 40% container flow empty (dep. On lines)**
- **45% of packaging material single use.**
- **4x usages for standard pallets.**
- **Manual customs processes.**
- **60% goods too fast.**
- **Huge inventory build ups.**

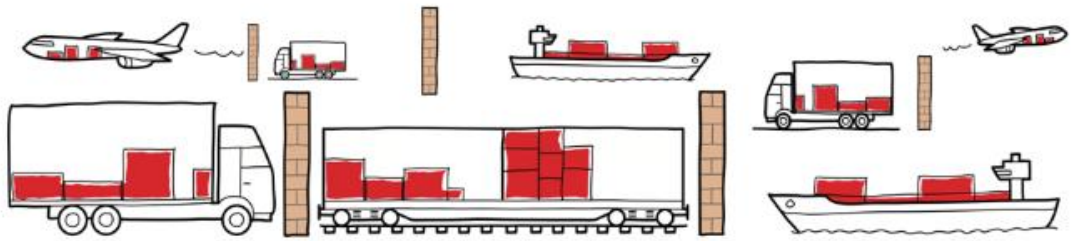
The Physical Internet:



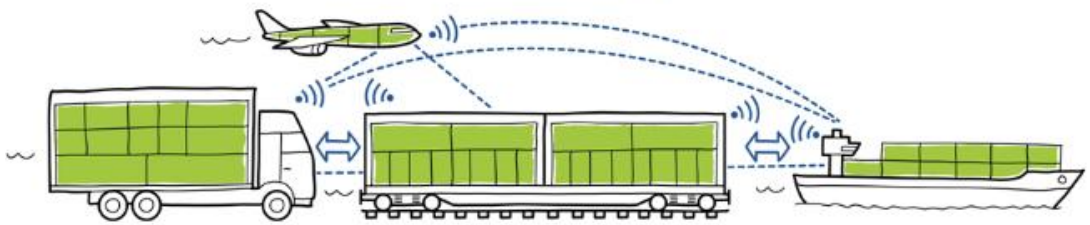


[The Physical Internet | Physical Internet Center]

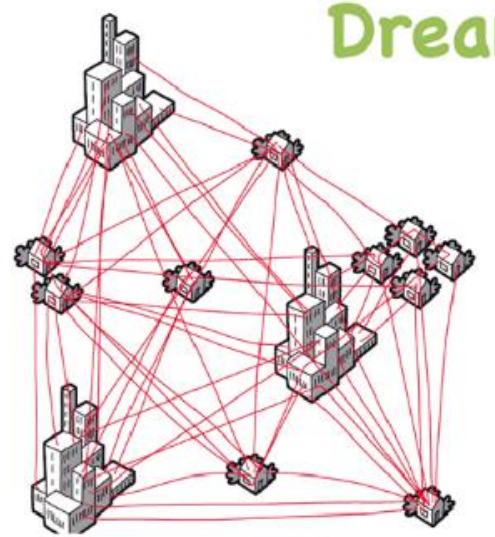


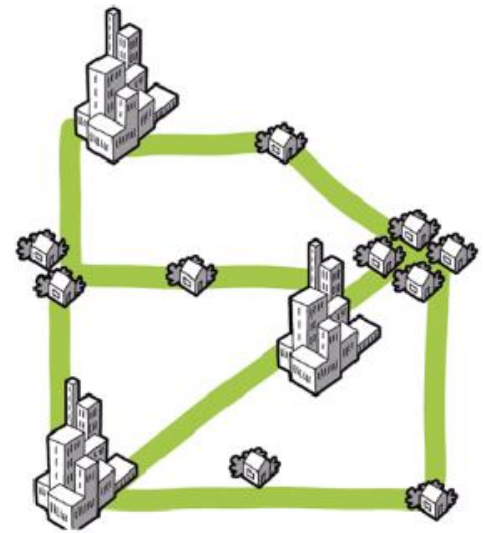


Challenge



Dream





How to Technically achieve your solution?



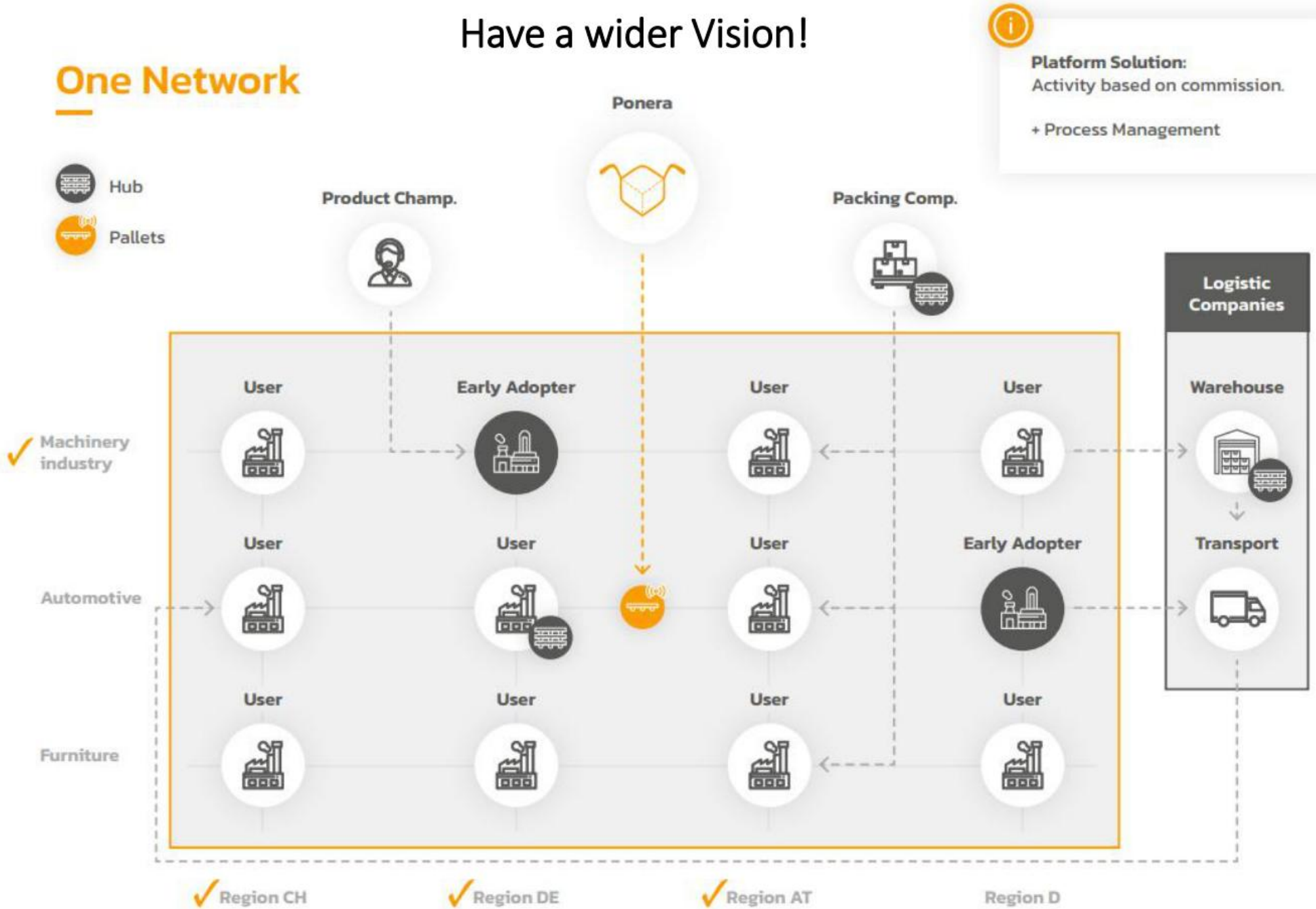


?



Have a wider Vision!

One Network



You know what problem(s) you want to solve, now what?

Top Reasons for Startup Failure

1



No Market Need

42%

2



Running Out of Cash

29–38%

3



Not the Right Team

23%

4



Getting Outcompeted

19%

5



Pricing and Cost Issues

18%

6



Poor Product & Ignoring Customers

17% / 14%

7



Poor Marketing

14%

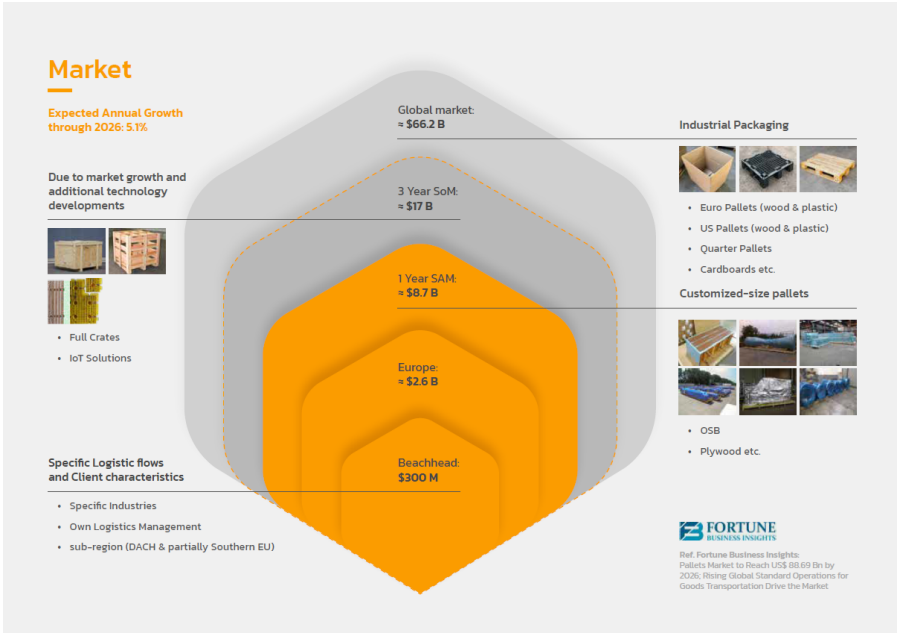
8



Co-founder Disharmony / Investor Mismatch

13%





Cheapest Route to Finding Bottlenecks

Ponera Positioning in Competitive Landascape



Bubble size shows the carbon footprint, the bigger, the higher the footprint

1. PONERA
2. CHEP
3. KIFA, DEUFOL, dpl Weinzierl Verpackungen GmbH, Sarl Chassignol Charlin, cargopack tägi ag, NEFAB SAVING RESOURCES
4. utz, MAUER PACKAGING
5. EPAL
6. LESER, BRIEGER VERPACKUNGEN
7. TURTLE, ROKBOX

International Physical Internet Conference 2019



Talk to the Market.. even early on, when your just a concept.

Developing the Solution.

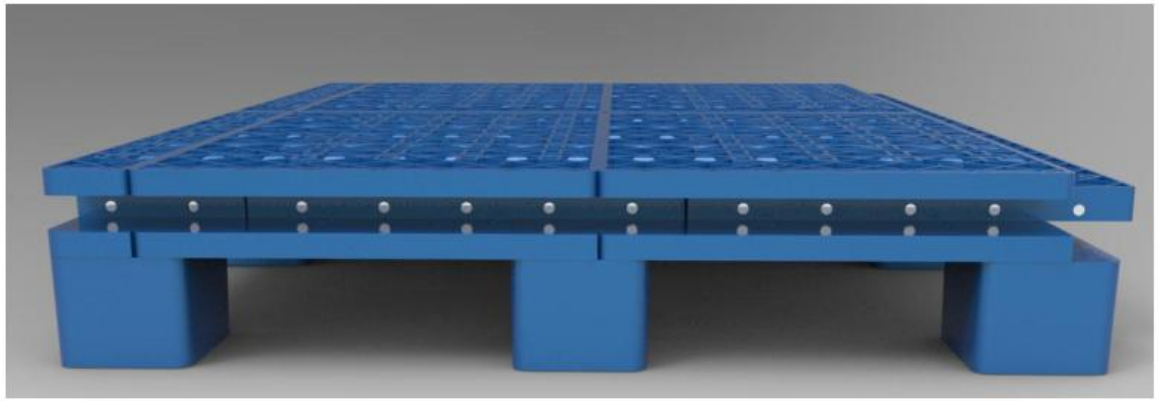
Minimal Viable Product:

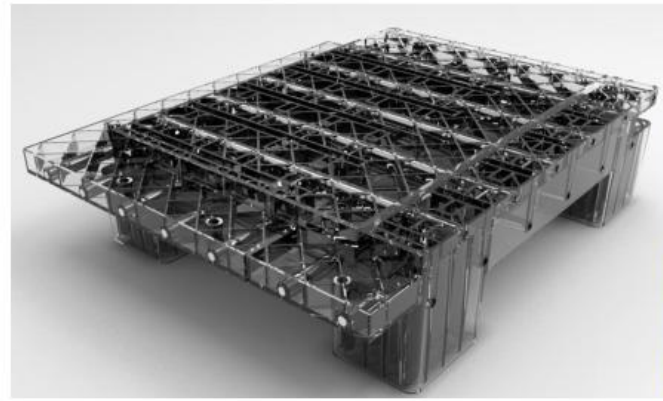
- The simplest version of the product that delivers real value.

1. Validate early: Test whether the product solves a real customer problem and whether customers will pay.
2. Reduce risk and cost: Identify wrong assumptions before investing heavily in development.
3. Learn and iterate: Gather real customer feedback to improve the proposed solution.

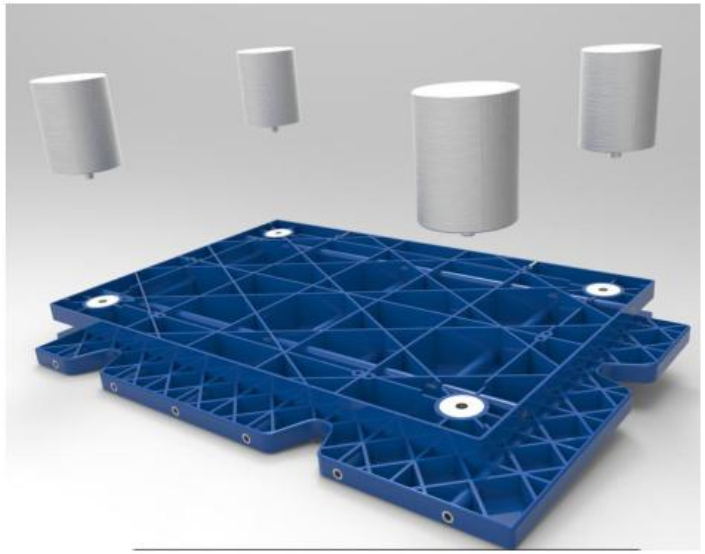


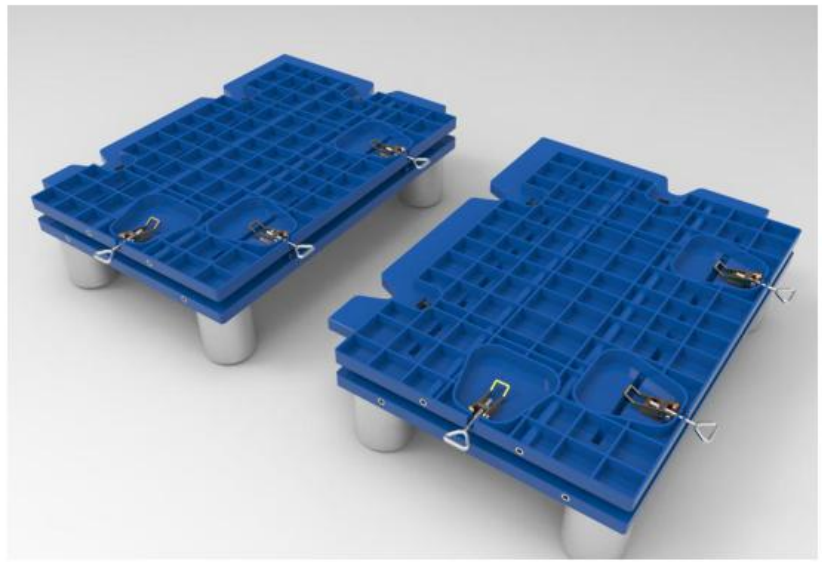
Early Designs and Prototypes:













First Milestones: Validate, De-risk, and Secure Financing

Finance milestones, not an abstract five-year dream

1 What to show in a pitch

Problem
Real and significant customer pain

Solution
Your approach and key differentiation

Market
Size, customers, willingness to pay

Team
Relevant skills and execution capability

Funding need
Clear amount and use of funds

Milestones
What will be achieved with this round

2 Reduce scope to reach first milestone

Separate the pallet from the full crate to de-risk and lower the capital requirement.

Full crate system



Investment:
several million CHF

- Multiple components
- More complex tooling
- Higher technical and industrial risk
- Longer time to market

One pallet module



Investment:
~300k CHF

- Focused development
- Simpler tooling
- Lower risk and cost
- Faster time to market
- Allows customer testing and early revenue

3 Key conditions to de-risk financing

Grant financing
Non-dilutive funding to de-risk early development

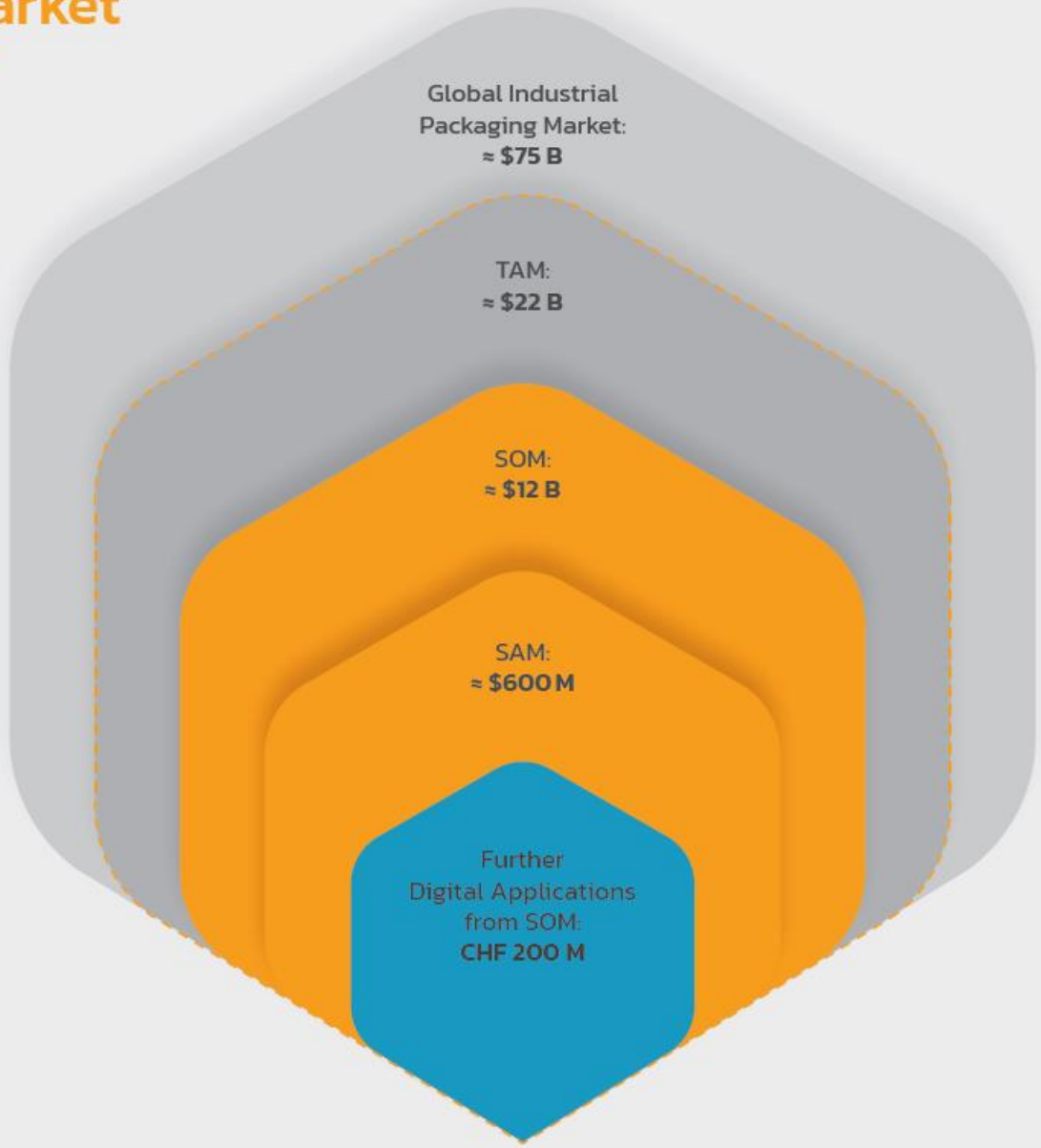
Signed LOI with clients
Customers committed to participate in development costs (e.g. pilots, tooling)

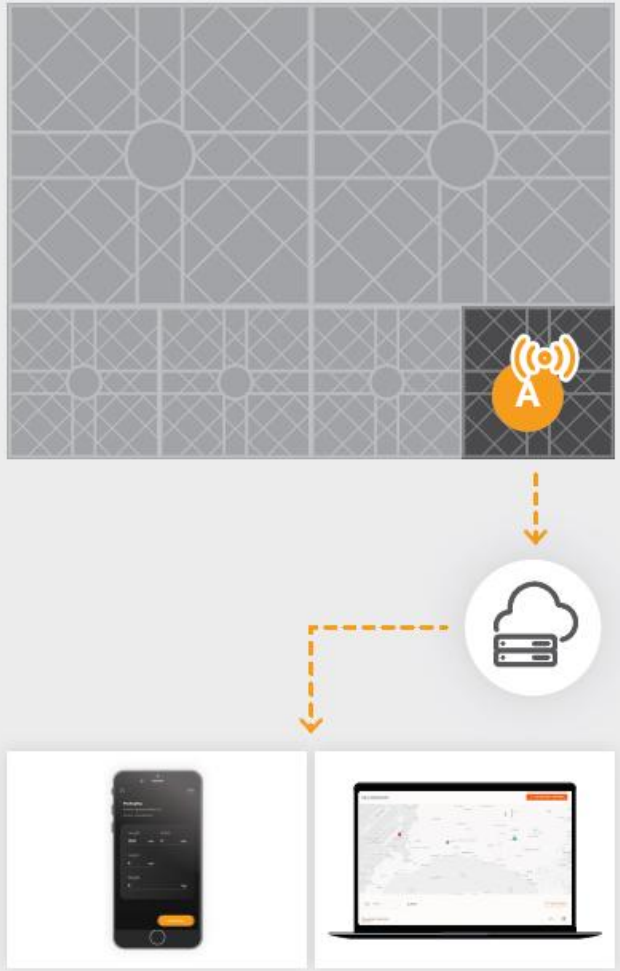
Clear, fundable milestones
Show concrete next step with measurable outcome

Funding needed

Market Validation 500 K CHF Target (1 M CHF Acheived)	**Complete Solution** 1.5 M CHF + 1.5 M CHF	**Scale** 5 M CHF
Investments: Investment required to create the product (R&D) Net Working Capital for Pilot activities Business devel. activites Multiple international patents and national entries filed	**Investments:** Supply Chain Build Up Further R&D on Hardware & IoT Solutions Distribution Scaling (Business Dev)	**Investments:** Develop the renting model Improve the IOT Solutions Business development activi-ties
Outcomes: Pilot Activity completed Validated market appetite MVP for small projects SC Build experience	**Outcomes:** Production and Supply Chain in place Complete crate solution Sales organization	**Outcomes:** Renting Model in place (Mix Sales/Rental) IOT Solutions

Market



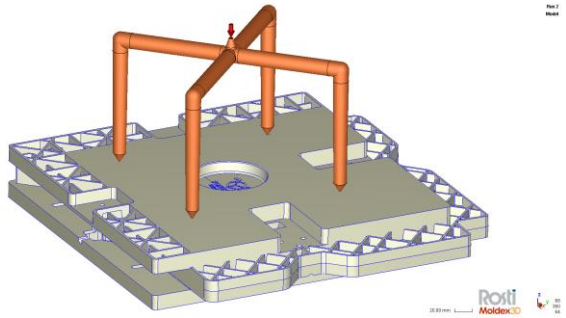


FORTUNE BUSINESS INSIGHTS

Ref. Fortune Business Insights:
Pallets Market to Reach US$ 88.69 Bn by 2026; Rising Global Standard Operations for Goods Transportation Drive the Market

GENERAL INFORMATION	
Customer	Ponera
Project Title	Roof
Part Name	model_Run1.mfe
Analysis Variation	Run02
Analysis Goal	FPW
Analysis Ref Number	MDX_0536

PART 3D



MATERIAL INFORMATION			
Polymer Type	PP		
Producer	N/A		
Trade Name	Celstran PP-GF30-02		
Melt Temp. (°C)	Min.	Norm.	Max.
	200	210	220
Mold Temp. (°C)	Min.	Norm.	Max.
	65	70	75
Ejection Temp. (°C)	90		
Freeze Temp. (°C)	110		

ANALYSIS SETUP CONDITIONS	
Melt Temp. (°C)	210
Mold Temp. (°C)	70
Injection Time (s)	3
Packing Time (s)	50
Cooling Time (s)	27.8
V/P Switchover %	96
Injection Volume (cc)	2270.56
Mesh Type	3 BLM 3D
Mesh Element Size	2

PROCESS/ACTUAL RESULTS			
Fill (s)	**Pack (s)**	**Cool (s)**	**Open (s)**
3.08	53.5	-	-
Cycle Time (s)		-	
RESULT SUMMARY			
Sprue Pressure (MPa)		20.187	
Max Clamp Force (t)		Fill	172.388
		Pack	236.607
Shot Weight (g)		2209.840	
XYZ Distortion (mm)		0.05 ~ 1.07	

Production Tooling – Each component is a new project








09.2020-12.2020:
Production Capabilities


01.2021:
Modules
Produced


02.2021:
Industrial Operations

Ready for Market?



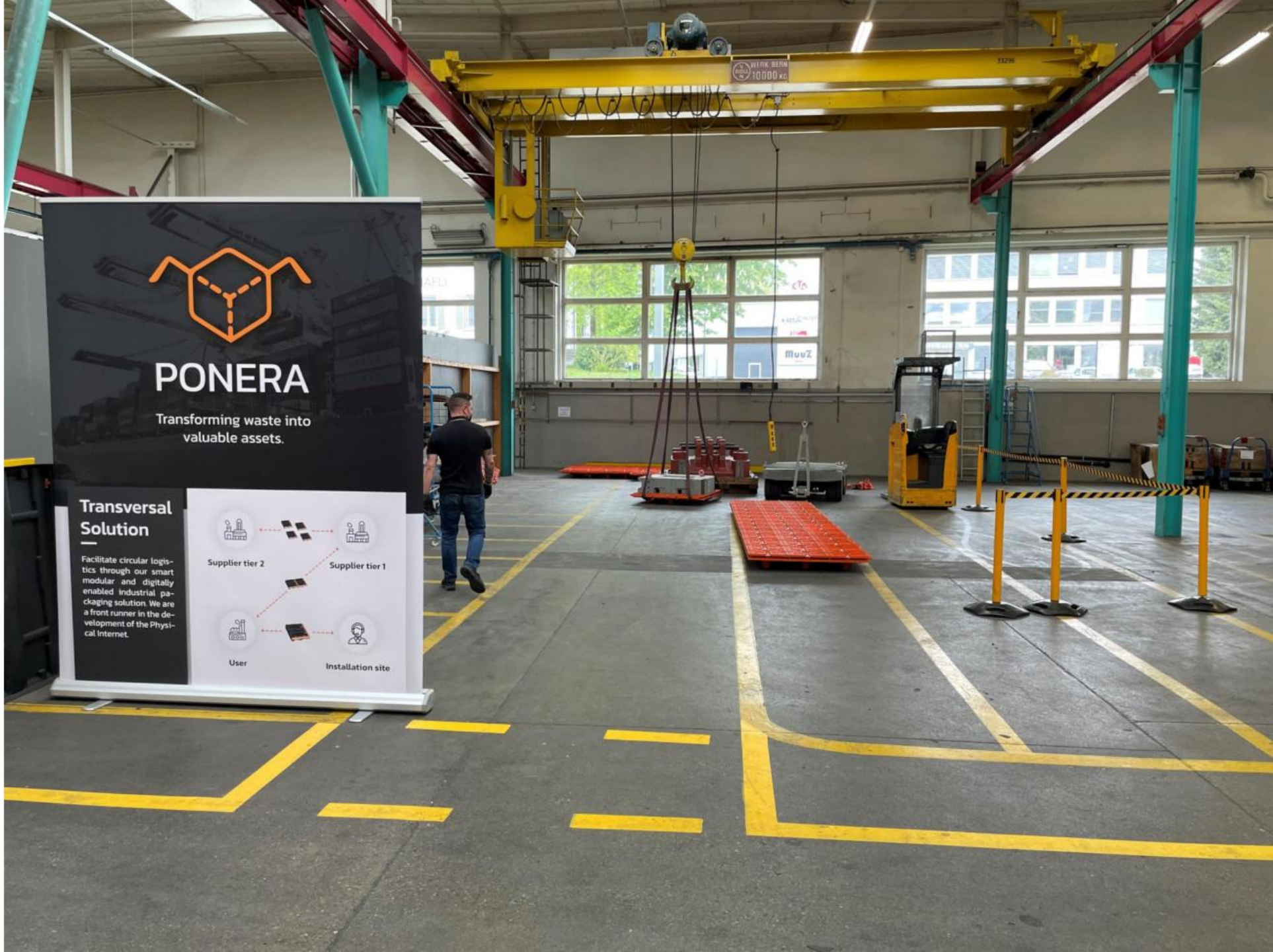
After Multiple Iterations..



Testing Product Readiness in Real Operations

- Shopfloor Testing
- SOP improvement
- Milkrun Switzerland
- Customer Projects

GO/NOGO decision



POC review







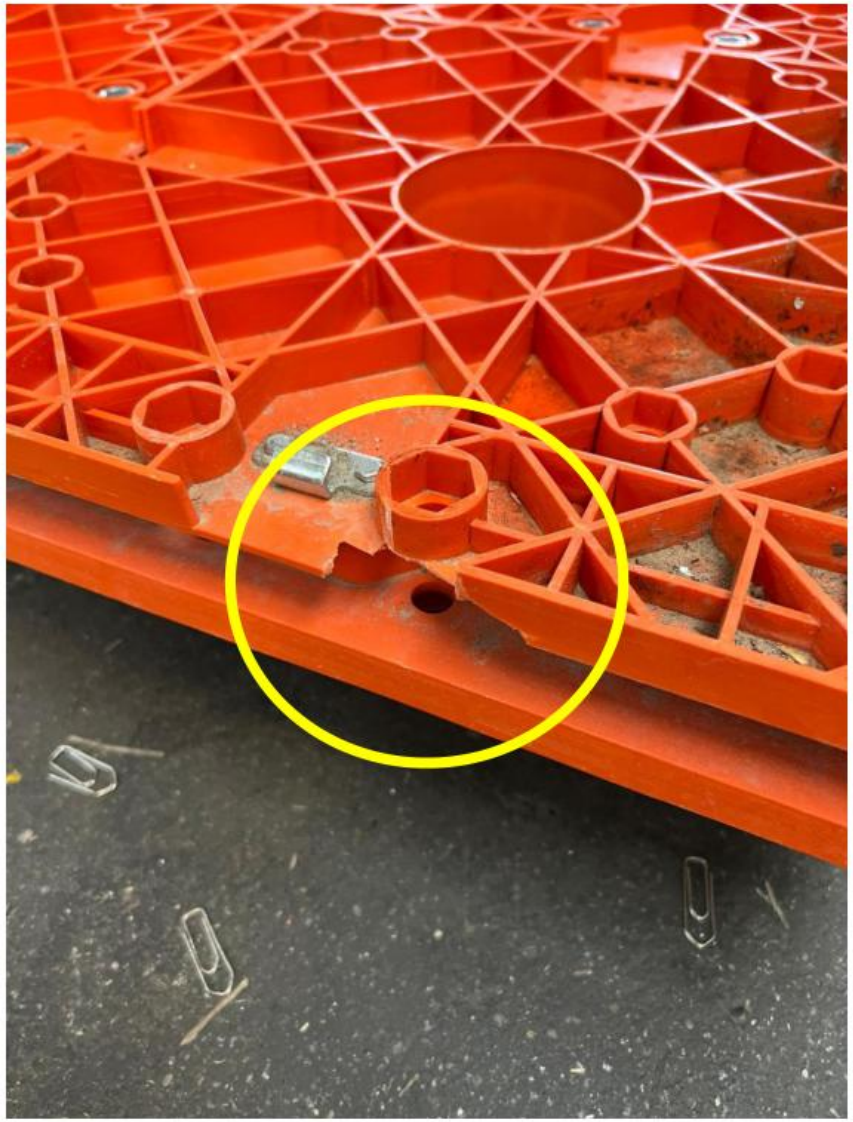


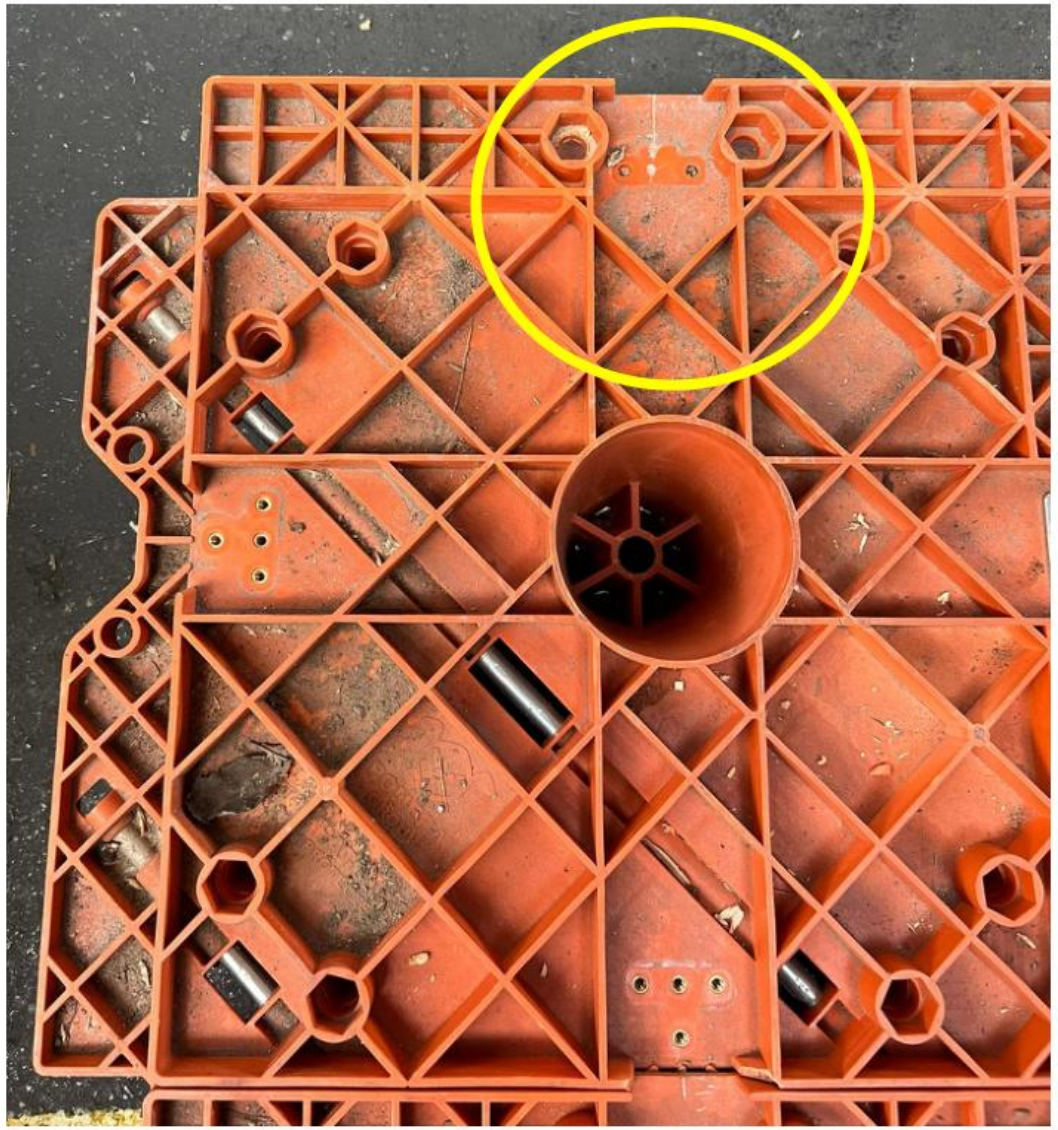
Different clients, using different process, diverse requirements..

More improvements needed..



Requalification?





Ponera Solution Reusable Modular Industrial Packaging





> 15 Tons

How to Actually Sell Innovation

Reduce the customer's perceived risk and make adoption as easy as possible

1 Process



- Understand customer decision process
- Identify key stakeholders
- Define clear next steps (pilot → evaluation → contract → scale)
- Align internal timelines with customer cycle

2 Relationships



- Build long-term trust
- Work with champions inside the organization
- Engage at multiple levels (technical, operations, commercial, management)
- Be present and supportive during testing and rollout

3 Business Model



- Make it easy to buy and use
- Flexible models (purchase, lease, subscription, pay-per-use)
- Clear and fast ROI
- Minimize upfront cost for the customer

4 Remove Adoption Barriers



- Compatibility with existing processes
- Minimal changes to infrastructure
- Support for integration and training
- Proven reliability and performance
- Address regulatory, operational and IT concerns

5 Create Urgency



- Clear and compelling value proposition (cost, efficiency, ESG, etc.)
- Time-limited incentives or pilot programs
- Use external drivers (market, regulation, competitive pressure)
- Show evidence from early adopters



Selling disruptive B2B products means reducing the customer's perceived risk.

- Lower risk to try
- Easy to implement
- Clear value and ROI

From pilot to long-term customer

Create and Improve a Replicable Sales System

Sales Cycle

1 - 3 Months
First contact to Pilot: Packers: Conversion rate: > 80 %
or
2 – 6 Months
First contact to Pilot: Users

1 - 2 Weeks
Case studies

1 - 3 Months
Try-vesting
or
3 Months
Pilot

1 - 3 Months
Analytics for implemetation

100% conversion rate to date

Reality vs. Plan?

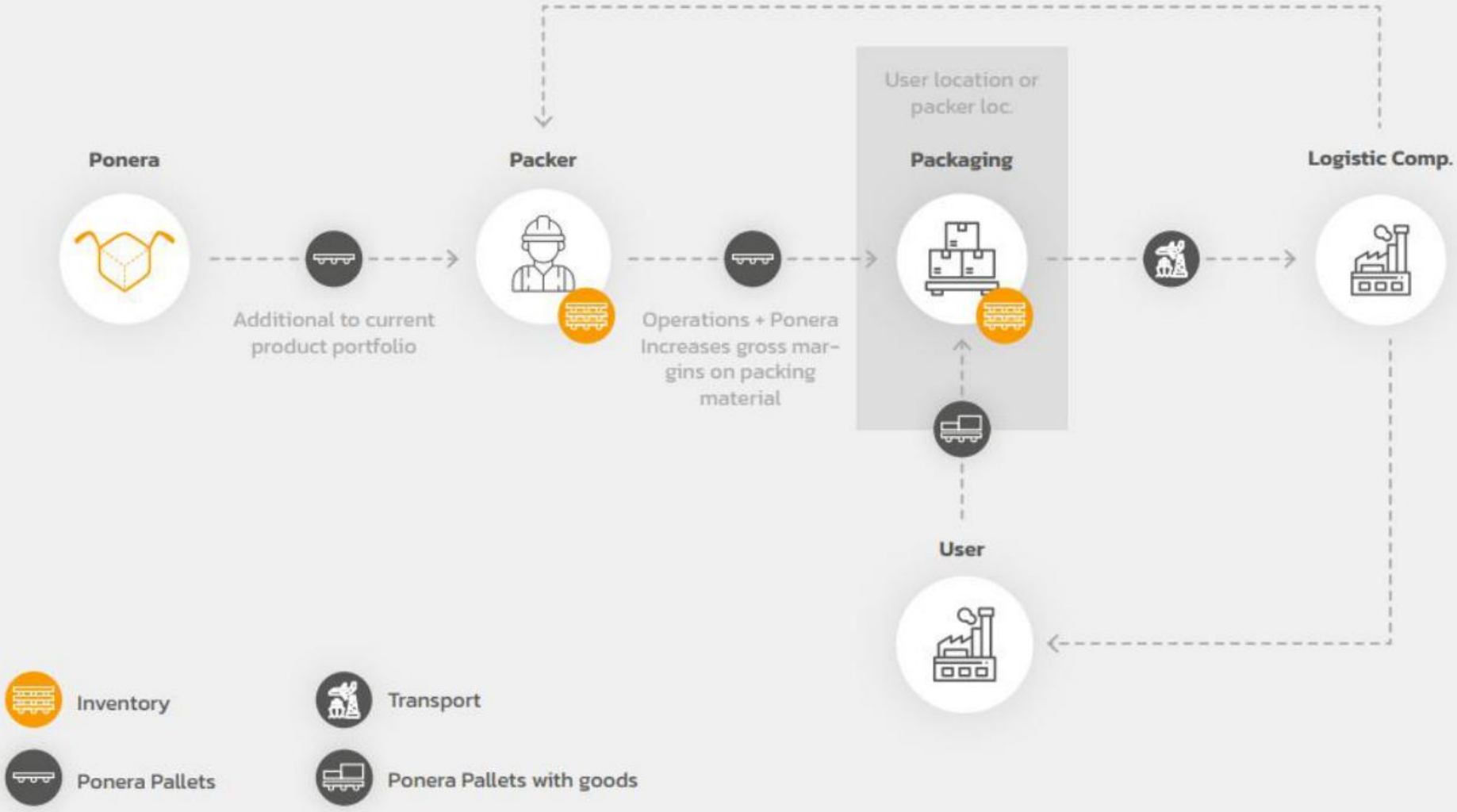
6 - 9 Months
Immediate impact

12 - 24 Months
Further implementation

12 - 24 Months
Analytics for implemetation

Confidential

Providing value to packing industry



With multiple stakeholders, who is the client? Defining client is as important business model itself.

You are not only selling a product!

Client G Case Study (CH - RO)

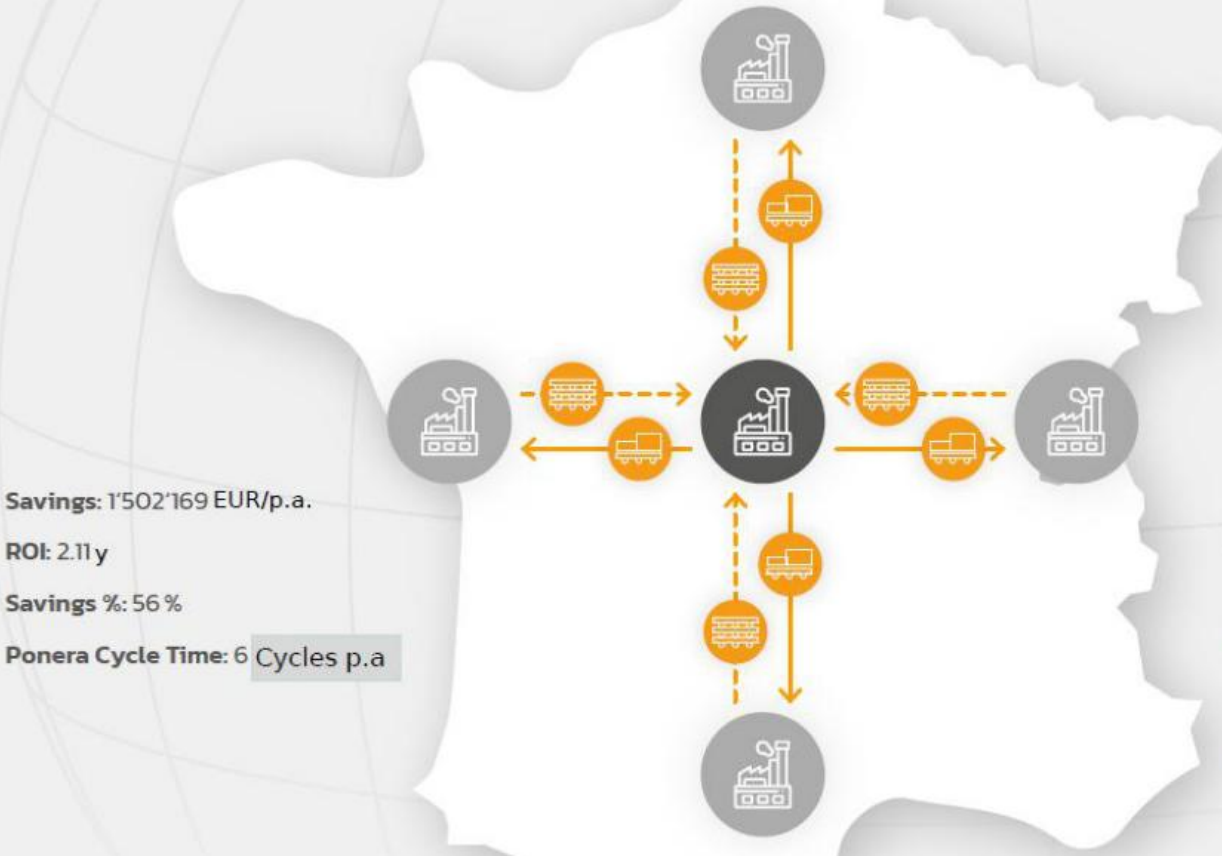


Route: TG (CH) – Bucarest (RO)	
Volumes: 2025 + 50%	
Replaced material Cost (per HU, excl. disposal): based on provided table 2023 (no inflation accounted).	
Reverse Trucking Cost: CHF 2'000 per FTL	
Cycles per annum: 6	
Truck CO2: 64g /kmT	**Ship CO2:** 15g /kmT
Lease to Buy on below CF Example: 2 years, 10% down, 3% final purchase cost	
Depreciation: 6 years (conservative) & 10% replacement rate	

KPI	Year 1 Positive Cash Flow	Year 2 Positive Cash Flow	Year 3+ Positive Cash Flow	Savings (%)	Payback (years)	Wood Savings (tonnes p.a.)	CO2 Reduction (tonnes p.a.)
Figures	**CHF 67'882**	**CHF 80'829**	**CHF 155'293**	**73.0%**	**1.07**	**189**	**76**

- Support new process implementation
- Deliver measurable results.

Business Case FR – FR



Empty Modules

Loaded pallets/crates

Savings: 1'502'169 EUR/p.a.

ROI: 2.11 y

Savings %: 56 %

Ponera Cycle Time: 6 Cycles p.a

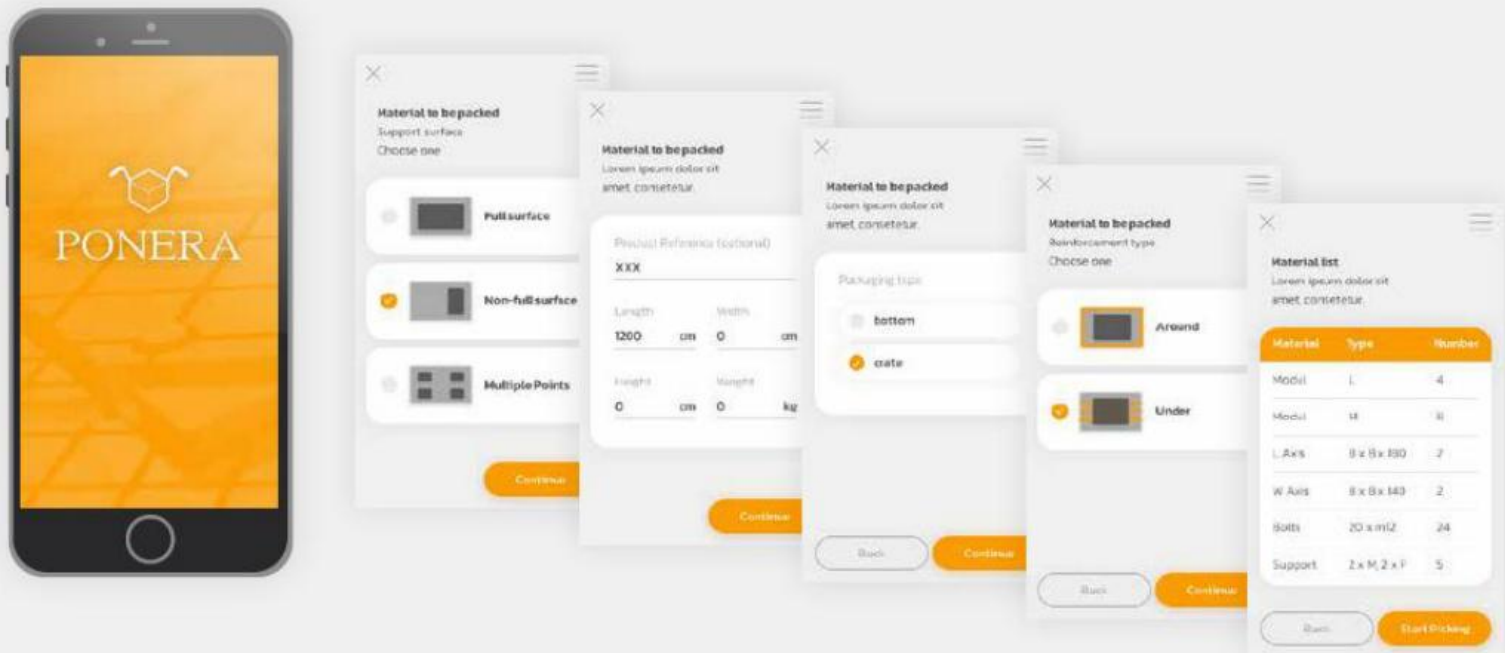
Assumptions:

- 42 delivery center each same volumes shipped (worse case)
- No added efficiency during loaded trasport (worse case)
- 2 months lay time (6 usages per year). With 6 weeks lay-time, average payback is below 1,5 years.
- Reverse costs EURO 1'500 per full truck (very high)

- Make implementation as seamless as possible.
- Eliminate hurdles/barriers.

Supporting our partners throughout the packing operations

- Guarantee easy and efficient packing operations.
- Guarantee full operator and transported material safety.
- Unprecedented transparency, standardization, and ability to optimize packing for users.



Market Intelligence to fine tune business model, can

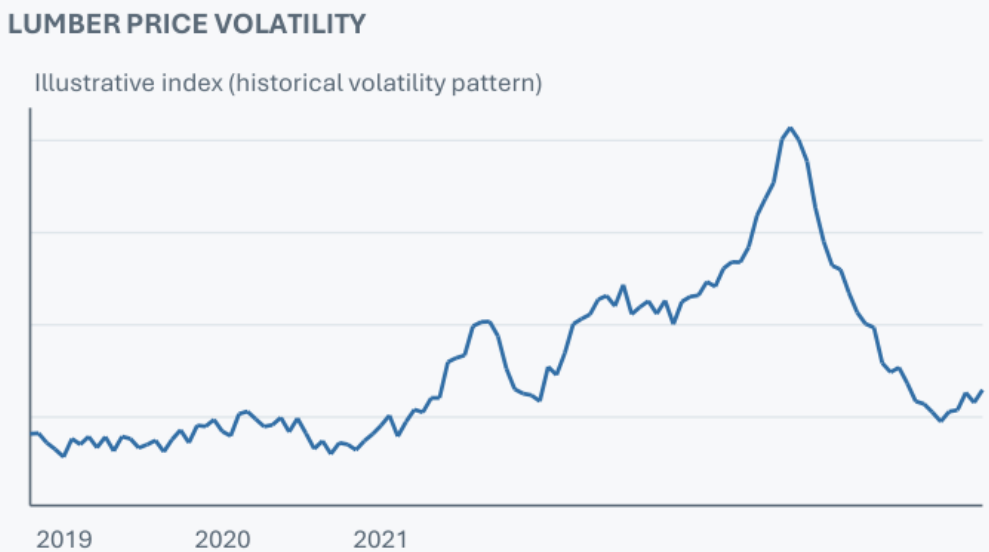
- Reduce customer barriers.
- Anticipate incentives.
- Provide optionality where it delivers added value.

Linear vs. Circular Business Models

Different risk profiles: variable purchasing costs vs. fixed assets and utilisation

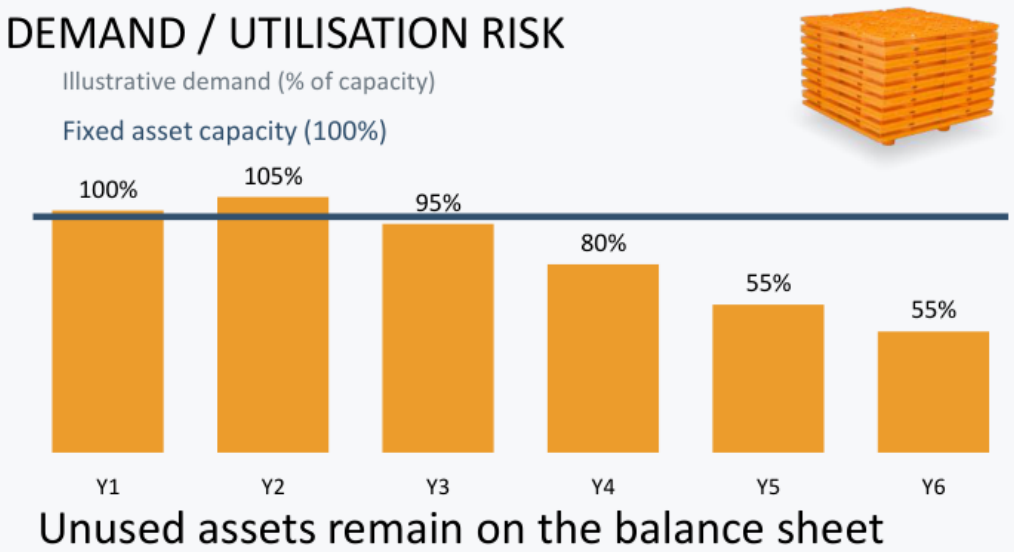
LINEAR | Buy as needed

+ Low upfront investment
– Volatile consumable prices; supply uncertainty



CIRCULAR | Own & reuse

+ More predictable costs; greater supply security
– Asset investment; future volume / demand risk

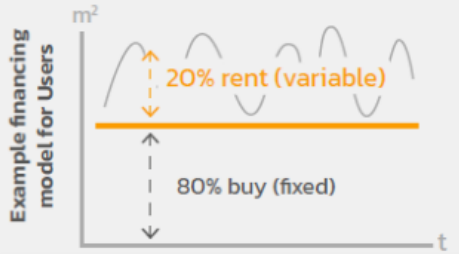


KEY TAKEAWAY **Circularity reduces price and supply risks, but business models must mitigate capital and utilisation risks to drive adoption.**

Business models enabled by key stakeholders.

- Approach:
  - Offer optionality.
  - Allow users to rent or pay-per-use, the product flexibly.
  - *Can require help of additional stakeholders for implementation.*
  - *Curve flattened with diversity in users.*

Product Development & Business Model



	1. Selling/Leasing Model	+ 2. Renting Model	+ 3. Activity Based Data Services
Business Model	≈ CHF 17 – 75.–/module Clients' Payback starting from 1 year	≈ CHF 0,05 – 0.20/module/day Decrease Client CAPEX Increase asset efficiency	2023 ≈ CHF 120 – 750.– subscription per User + ≈ CHF 1.–/module/year for activity based services Transparency / Analytics Process efficiencies
Product Development	• Heavy duty • Modular packaging system • Circular logistics analytics	• Additional auxiliaries • Platform phase 1 • Standard duty modular Pack. line	• Platform phase 2

Ponera Modular Boxes – Example Applications



Load : 3 ton
Price: **32,50 € /Month**



Load : 2,5 ton
Price: **11,50 € /Month**



Load : 600 kg
Price: **7,60 € /Month**



Load : 250 kg
Price: **6,70 € /Month**



Load : 400 kg
Price: **9,60 € /Month**



Load : 750 kg
Price: **16,50 € /Month**

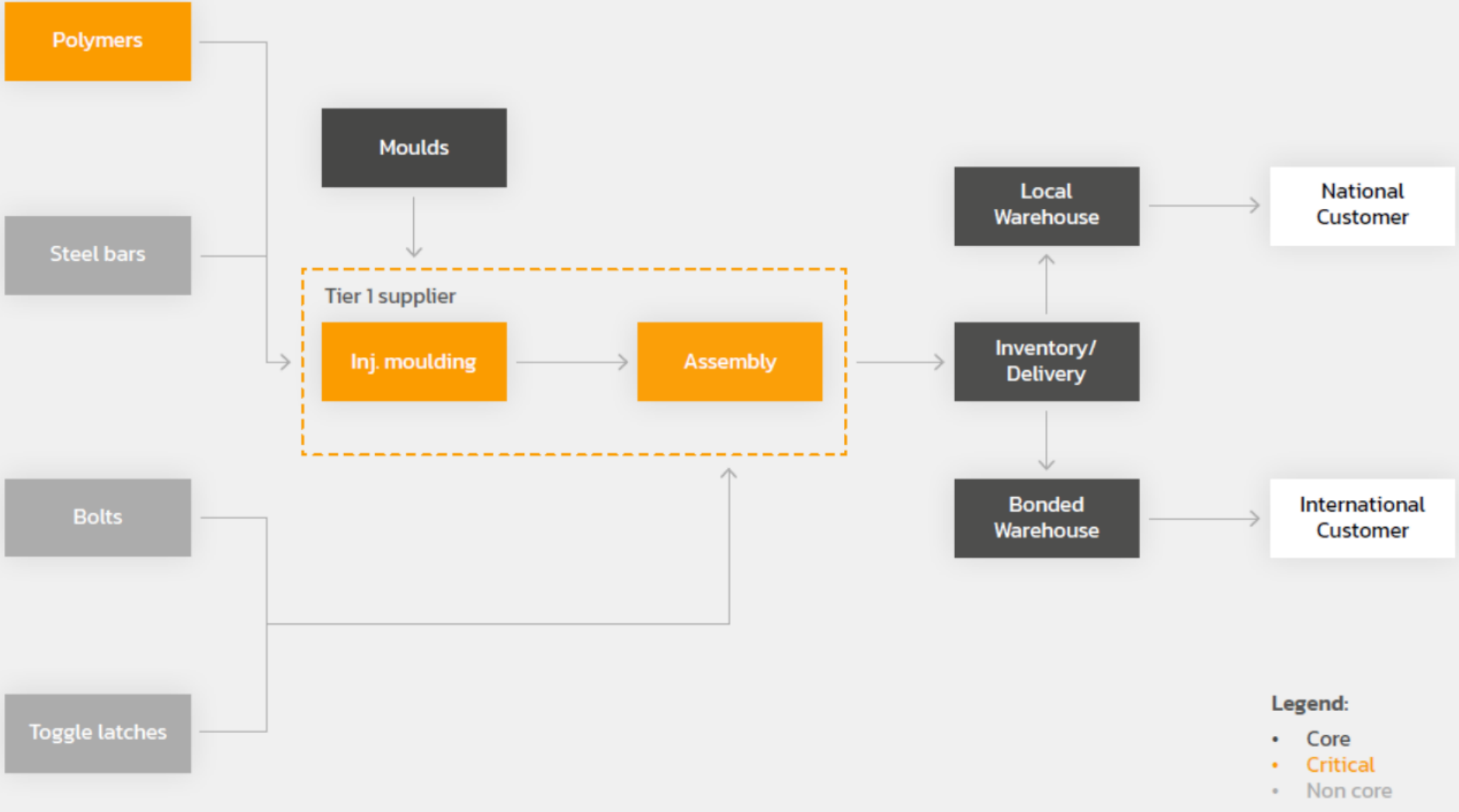

European Commission
EU NET-ZERO INDUSTRY ACT:
MAKING EUROPE THE HOME
OF CLEAN TECH
#NetZeroEU #NetZeroIndustry

ONERA

Funding needed

Market Validation 500 K CHF Target (1 M CHF Acheived)	Complete Solution 1.5 M CHF + 1.5 M CHF	Scale 5 M CHF
Investments: Investment required to create the product (R&D) Net Working Capital for Pilot activities Business devel. activites Multiple international patents and national entries filed	**Investments:** Supply Chain Build Up Further R&D on Hardware & IoT Solutions Distribution Scaling (Business Dev)	**Investments:** Develop the renting model Improve the IOT Solutions Business development activities
Outcomes: Pilot Activity completed Validated market appetite MVP for small projects SC Build experience	**Outcomes:** Production and Supply Chain in place Complete crate solution Sales organization	**Outcomes:** Renting Model in place (Mix Sales/Rental) IOT Solutions

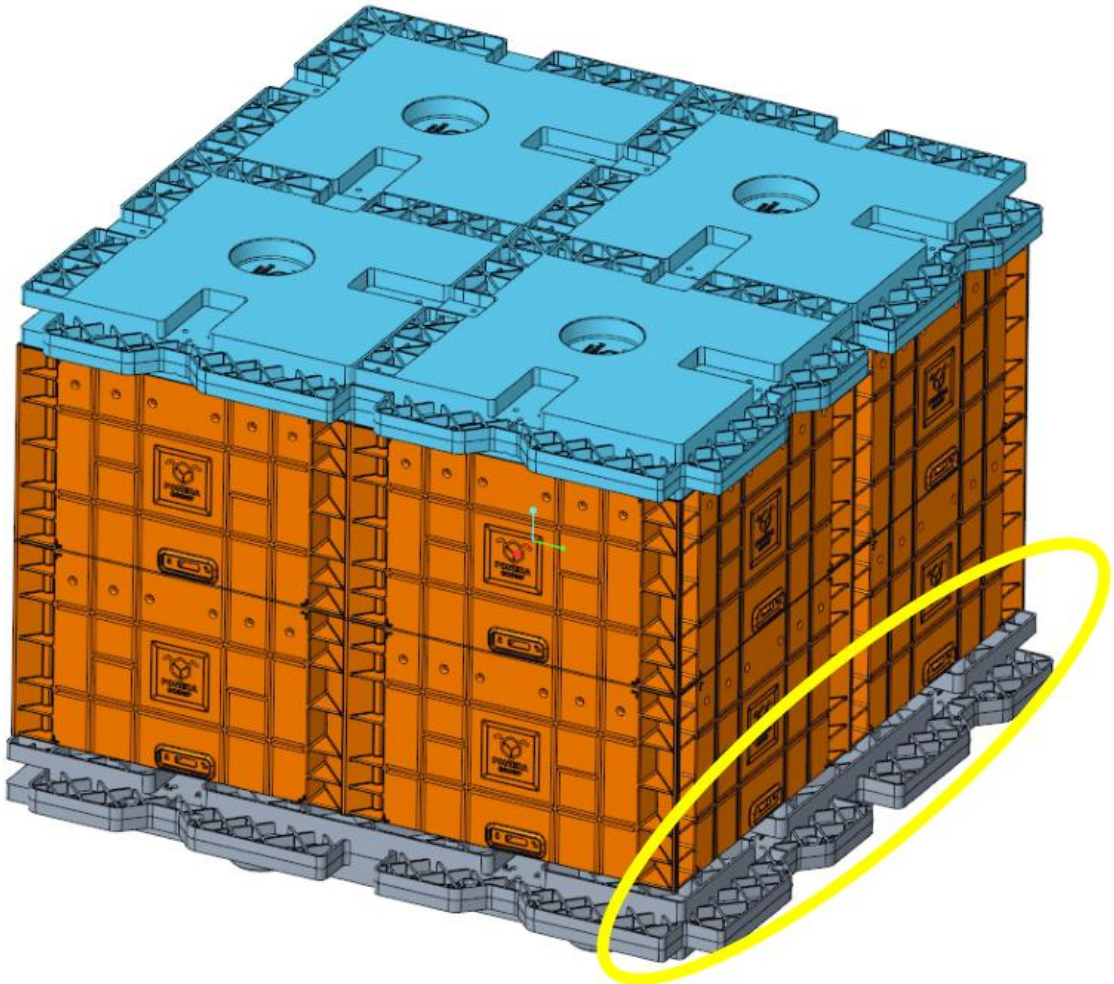
Structure – Supply Chain

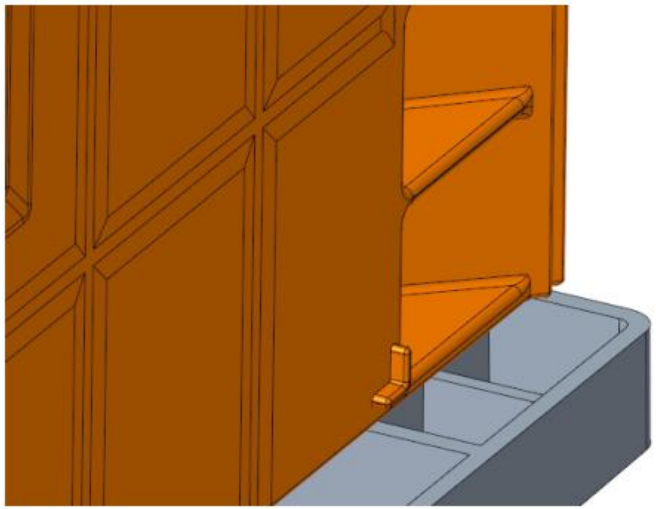


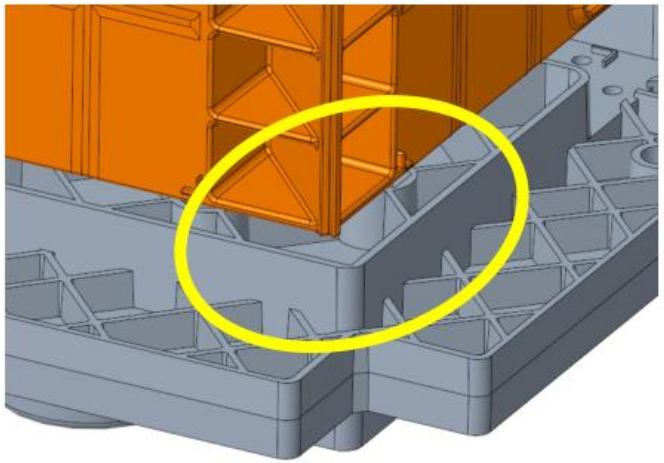
Production Capacity:

- **5 proprietary molds**
- **25k parts / month each**
- **International partners for expansion.**
- **+ Opened EU WH, with EORI number, Feb 2025.**

New Product Development, New Iteration Cycle, Catch Issues Early!














Prototyping



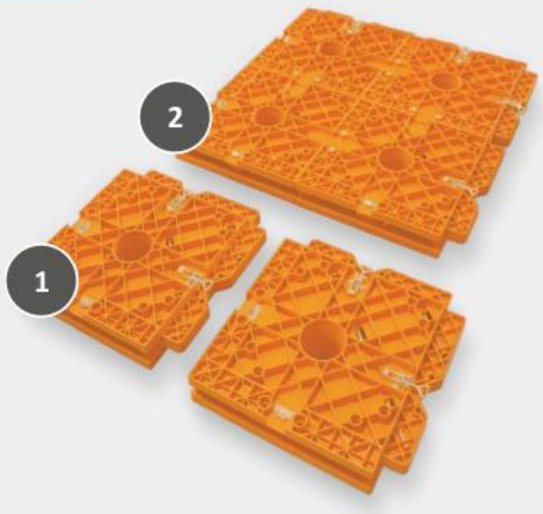




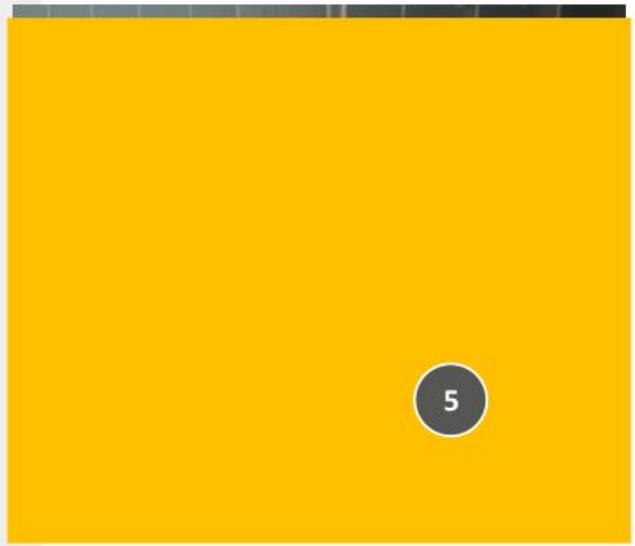
POC, Pilots and Roll-out

(with additional continuous improvement iterations)

Our products & Technical data







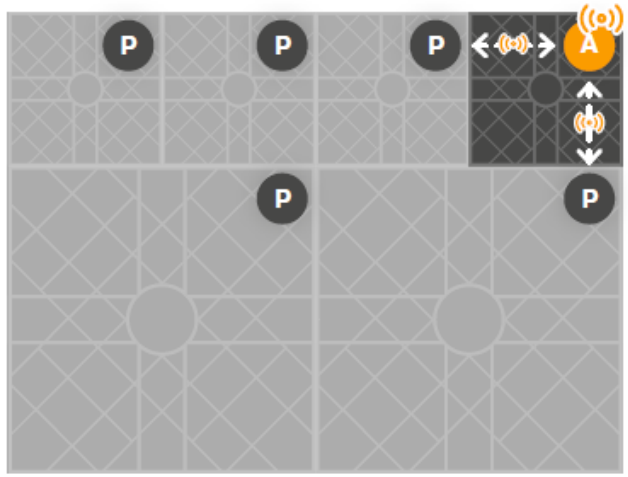
	Product	External dimension length (mm)	External dimension width (mm)	Weight (kg)	Static Load Capacity
1	Bottom Module Size M	375 + 45	375 + 45	2,75	3500 kg/m2
2	Bottom Module Size L	750 + 45	750 + 45	8,5	
3	Top Module	375	375 + 45	2,1	2000 kg/m2
4	Wall Module	375	35	0,8	

5 Number of additional improvements planned, after first customer projects.

Confidential

Ponera Brings Packaging to the Industry 4.0 Era

Digital IoT:



Server (Cloud)

Digital Software:



Ponera Mobile App:

- Seamlessly guiding users through Ponera process (SOP)
- Offline enablement
- 5 main product features



Ponera Platform:
Full transparency and real-time visualization of pallets, goods, and main data points

Circularity & Pooling:
Optimizing module fleet management

2 features:

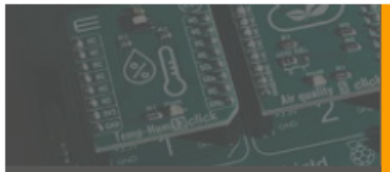
- Inventory Management
- Sharing/transaction enablement

Data Analytics:
Optimizing processes and operations directly or indirectly linked to packing/ freight transport.

- Algos, AI, Data Mining

Integration with other platforms

data encription



Qualitative Sensors
Temperature, Shock, Humidity, offline memory

Highly Cost effective
Multi-use, low handling, state of the art, combined technologies

Traceable (IoT)
GPS Geoloaction

Multiple Patent App.
International Utility patents

Governance: The Hardest Part

The hardest problems eventually become human rather than technical

1 Cofounders
Highest impact risk



- Aligned goals and long-term commitment
- Evaluate professional background and circumstances
- **Personality fit** (not friendship)
- Shared values and working style
- Clear roles, equity and decision rights

2 Employees



- Hire for key gaps and next milestones
- Incentives aligned with value creation (e.g. equity)
- Clear responsibilities and accountability
- Culture and values matter

3 Board



- Relevant experience and networks
- Support and challenge management
- Clear governance and decision processes
- Independent view, not just approval

4 Investors



- Aligned goals and time horizon
- Realistic expectations and level of involvement
- Value beyond capital (network, expertise)
- Manage different investor types and incentives

5 Incentives



- Align compensation with company and individual goals
- Equity, vesting and performance metrics
- Handle changes in contribution over time
- Be clear on what happens if someone leaves or stops contributi



"Companies are killed from within."

Conflicts, misaligned goals and governance issues — especially between cofounders — can destroy a company, often more than market or technical challenges.



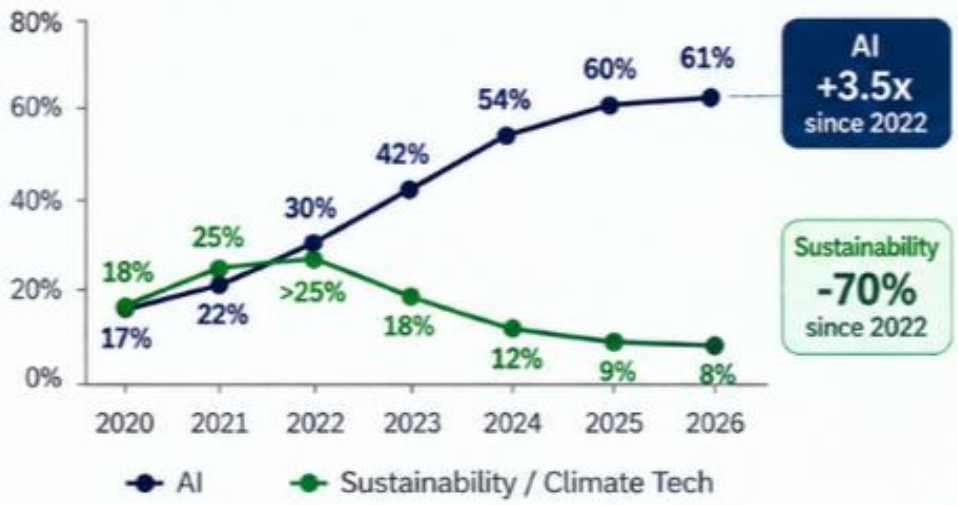
Get the right people, aligned goals, incentives and values — early.

Changes While You Execute

Investment sentiment, valuations, freight rates and regulation can shift significantly during the build-up phase.

Sustainability vs. AI: Share of Global Investment

% of total VC + PE investment

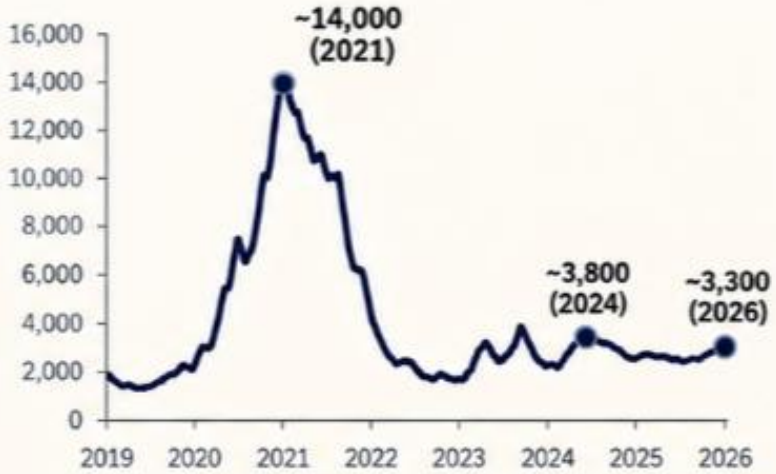


- Sustainability investment share peaked at >25% in 2022, now ~8% in 2026 (more than 70% decline).
- AI has increased from 30% in 2022 to ~61% in 2026.
- Capital, talent and valuations have shifted strongly toward AI, making it harder to raise large rounds for pure sustainability plays.

Source: PwC State of Climate Tech 2023 & 2024; OECD Global VC Trends 2025.

Global Freight Rates

Container spot rates (USD/40ft, Shanghai–Rotterdam)



- Freight rates surged in 2021, then normalized and remain much lower today.
- Lower logistics costs reduce near-term urgency for supply chain innovations.
- Market conditions can change quickly and impact customer ROI and adoption timing.

Source: Freightos Baltic Index (FBX) Global Container Index, monthly average.

EU Regulation: Single-Use Packaging (Industrial Products)

Recycled content targets for plastic transport packaging (pallets, crates and similar)

2022	Commission proposal: initial targets and draft rules.
2023–2024	Political negotiations and industry pushback.
2025	Further revisions and extended transition periods.
2030	**Recycled content requirements apply from 2030[1].**



- Single-use plastic transport packaging (e.g. pallets, crates) will be subject to recycled content targets from 2030.
- Timeline has been extended and revised multiple times during the legislative process.
- Do not build a business case that requires regulation to be profitable.

Source: EU Packaging and Packaging Waste Regulation (PPWR), Council/Parliament updates 2022–2024.

Never build a business case that requires regulation or favorable capital markets.

Investment cycles, freight rates, valuations and regulation can change — design for resilience and real customer value.

Main Lessons

What to remember for your own journey

1 

Start with a problem or find the right use case.

Problem-first or technology-first: both work.

2 

Choose people carefully

Align goals, roles and incentives. Evaluate competencies, fit and values (not just friendship).

3 

Validate early

Market, customer need, competition and FTO. Test real demand and freedom to operate.

4 

Move fast and iterate.

Prototype, test and learn. Perfection is the enemy of progress.

5 

Industrialization is a project of its own.

Different skills, time and capital required. Plan early.

6 

Finance milestones, not the five-year dream.

Cash is a project constraint. Secure capital for de-risking milestones.

7 

Customer interest is not customer validation.

Make adoption easy and low-risk. Don't underestimate problem urgency.

8 

Do not rely on regulation or favorable capital markets.

Plans, timelines and market sentiment can change significantly.

9 

Be flexible and adapt.

Expect uncertainty. Reassess assumptions and pivot when needed.